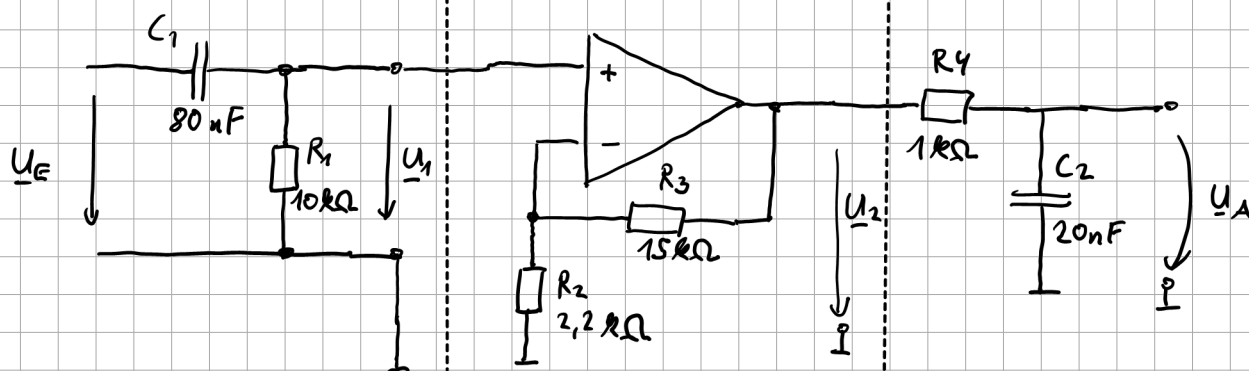


BOD 3



$$\underline{A}_1 = \frac{\underline{U}_1}{\underline{U}_E} = \frac{R_1}{R_1 + \underline{Z}_{C1}}$$

$$= \frac{R_1}{\frac{1}{j\omega C_1} + R_1} \cdot \frac{j\omega C_1}{j\omega C_1}$$

$$= \frac{j\omega \underbrace{C_1 R_1}_{\tau_1}}{1 + j\omega \underbrace{R_1 C_1}_{\tau_1}}$$

$$\tau_1 = \frac{1}{\omega_1}$$

$$\omega_1 = 2\pi f_1$$

$$\underline{A}_1 = \frac{j \frac{\omega}{\omega_1}}{1 + j \frac{\omega}{\omega_1}}$$

$$= j \frac{\frac{2\pi f}{2\pi f_1}}{1 + j \frac{2\pi f}{2\pi f_1}}$$

$$1 + j \frac{2\pi f}{2\pi f_1}$$

Hochpass

$$\underline{A}_1 = \frac{j \frac{f}{f_1}}{1 + j \frac{f}{f_1}}$$

$$\tau_1 = R_1 \cdot C_1$$

$$= 10 \text{ k}\Omega \cdot 80 \text{ nF}$$

$$= 800 \mu\text{s}$$

$$\omega_1 = \frac{1}{\tau_1} = 1250 \frac{1}{\text{s}}$$

$$\underline{f_1} = \frac{\omega_1}{2\pi} = \underline{199 \text{ Hz}}$$

$$\underline{A}_2 = \frac{\underline{U}_2}{\underline{U}_1} = 1 + \frac{R_3}{R_2}$$

$$= 1 + \frac{15 \text{ k}\Omega}{2.2 \text{ k}\Omega}$$

$$= 7.82$$

Konstante

$$K_{dB} = 20 \cdot \log 7.82$$

$$= \underline{17.9 \text{ dB}}$$

$$\underline{A}_3 = \frac{\underline{U}_A}{\underline{U}_2} = \frac{\underline{Z}_{C2}}{R_4 + \underline{Z}_{C2}}$$

$$= \frac{\frac{1}{j\omega C_2}}{R_4 + \frac{1}{j\omega C_2}} \cdot \frac{j\omega C_2}{j\omega C_2}$$

$$= \frac{1}{1 + j\omega \underbrace{R_4 C_2}_{\tau_2}}$$

$$\tau_2 = \frac{1}{\omega_2} = \frac{1}{2\pi f_2}$$

$$= \frac{1}{1 + j \frac{\omega}{\omega_2}}$$

$$= \frac{1}{1 + j \frac{2\pi f}{2\pi f_2}}$$

$$\underline{A}_3 = \frac{1}{1 + j \frac{f}{f_2}} \quad \text{Tiefpass}$$

$$\tau_2 = 1 \text{ k}\Omega \cdot 20 \text{ nF} = 20 \mu\text{s}$$

$$\omega_2 = \frac{1}{\tau_2} = 50 \text{ k} \cdot \frac{1}{\text{s}}$$

$$\underline{f_2} = \frac{\omega_2}{2\pi} = 7958 \text{ Hz}$$

$$\underline{\approx 8 \text{ kHz}}$$

Aufgabe: Hochpass Konstante Tiefpass

$$A_{ges} = \frac{j \frac{f}{f_1}}{1 + j \frac{f}{f_1}} \cdot 7,81 \cdot \frac{1}{1 + j \frac{f}{f_2}}$$

$f_1 = 200 \text{ Hz}$
 $f_2 = 8 \text{ kHz}$
 $K_{dB} = 20 \cdot \log_{10}(7,81) = 17,9 \text{ dB}$

